

## 6. Normal subgroups, quotients, and extensions

A homomorphism can send different group elements to the same image. The elements sent to the identity form its kernel. Each coset of the kernel consists of all the elements with one particular image. This chapter shows that these cosets form a group isomorphic to the image. We then ask how a group can be constructed from a given normal subgroup and quotient. Rotations and translations, glide reflections, and operators carrying phases illustrate these constructions.

### 6.1. NORMAL SUBGROUPS

READING. [Moore, §7.3, pp. 80–92].

Conjugation carries a subgroup  $H$  to another subgroup  $gHg^{-1}$ . Some subgroups are carried onto themselves by every element of  $G$ .

normal subgroup  
正规子群

**Definition 6.1** (Normal subgroup). A subgroup  $N \subseteq G$  is *normal* in  $G$ , written  $N \triangleleft G$ , if

$$gNg^{-1} = N \quad \forall g \in G.$$

This is equality of sets. It permits conjugation to permute the elements of  $N$ ; it does not require  $gng^{-1} = n$  for each  $n \in N$ . That stronger condition says  $N \subset Z(G)$ .

**Proposition 6.2.** *For a subgroup  $N \subset G$ , the following conditions are equivalent.*

- (i)  $N \triangleleft G$ ;
- (ii)  $gN = Ng \quad \forall g \in G$ ;
- (iii)  $N$  is a union of conjugacy classes of  $G$ .

*Proof.* Multiplying  $gNg^{-1} = N$  on the right by  $g$  gives (ii), and the same step reversed gives (i). Condition (i) implies that the whole conjugacy class of each  $n \in N$  lies in  $N$ , giving (iii). Conversely, (iii) gives  $gNg^{-1} \subset N$  for every  $g$ . Applying this inclusion to  $g^{-1}$  and conjugating by  $g$  gives the reverse inclusion.  $\square$

**Proposition 6.3** (Kernels are normal). *If  $\varphi : G \rightarrow G'$  is a homomorphism, then  $\ker \varphi \triangleleft G$ .*

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*Proof.* The kernel is a subgroup by §3.2. For  $k \in \ker \varphi$ ,

$$\varphi(gkg^{-1}) = \varphi(g)e'\varphi(g)^{-1} = e',$$

so  $g(\ker \varphi)g^{-1} \subset \ker \varphi$ . Applying the same argument to  $g^{-1}$  gives equality.  $\square$

*Example 6.4.* The subgroups  $\{e\}$  and  $G$  are always normal in  $G$ . Every subgroup of an Abelian group is normal, since  $ghg^{-1} = h$ . The centre  $Z(G)$  is normal even when  $G$  is not Abelian.

*The sign homomorphism.* Recall from Chapter 5 that  $\text{sgn} : S_n \rightarrow \{1, -1\}$  sends even permutations to 1 and odd permutations to  $-1$ , with multiplication as the group operation in  $\{1, -1\}$ . Its kernel consists of the even permutations, so

$$A_n = \ker(\text{sgn}) \triangleleft S_n.$$

*The determinant homomorphism.* For  $\kappa = \mathbb{R}$  or  $\mathbb{C}$ , the multiplicative property  $\det(AB) = \det A \det B$  makes

$$\det : GL(n, \kappa) \longrightarrow \kappa^*$$

a homomorphism, where  $\kappa^*$  is the group of nonzero scalars under multiplication. Its identity is 1, so the kernel consists of the matrices with determinant 1. Thus

$$SL(n, \kappa) = \ker(\det) \triangleleft GL(n, \kappa).$$

As seen in §2.5, the determinant takes values in  $\{1, -1\}$  on  $O(n)$  and in  $U(1)$  on  $U(n)$ . These restricted homomorphisms have kernels  $SO(n)$  and  $SU(n)$ , respectively, so

$$SO(n) \triangleleft O(n), \quad SU(n) \triangleleft U(n).$$

**Proposition 6.5.** *Every subgroup of index two is normal.*

*Proof.* If  $[G : H] = 2$  and  $g \notin H$ , then  $gH = G \setminus H = Hg$ : both left and right cosets partition  $G$  into  $H$  and its complement. For  $g \in H$ , both cosets equal  $H$ . Apply Proposition 6.2.  $\square$

*Remark 6.6* (The role of the normalizer). Definition 4.30 now has a shorter interpretation:

$$H \triangleleft N_G(H).$$

Moreover,  $N_G(H)$  is the largest subgroup of  $G$  in which  $H$  is normal. Indeed,  $H \triangleleft K \subset G$  implies  $kHk^{-1} = H$  for every  $k \in K$ , hence  $K \subset N_G(H)$ . In particular,  $H$  is normal in  $G$  precisely when  $N_G(H) = G$ .

**Definition 6.7** (Simple group). A nontrivial group  $G$  is *simple* if

$$\forall N \triangleleft G, \quad N = \{e\} \text{ or } N = G.$$

simple group  
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For a prime  $p$ ,  $\mathbb{Z}_p$  is simple by Lagrange's theorem. The group  $A_4$  is not simple: its identity and three double transpositions form the normal subgroup  $V_4$ . It is a subgroup by Example 5.16, and conjugation preserves cycle type. We state without proof the further result that  $A_n$  is simple for  $n \geq 5$ .

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**Quiz 6.1.** In  $S_3$ , show that  $\langle(123)\rangle$  is normal but  $\langle(12)\rangle$  is not. Does being Abelian imply that a subgroup is normal in the ambient group?

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## 6.2. QUOTIENT GROUPS

Chapter 4 used  $G/H$  as a *set* of left cosets. Normality is exactly what allows multiplication in  $G$  to descend to multiplication of these cosets.

quotient group  
商群

**Theorem 6.8** (Quotient group). *If  $N \triangleleft G$ , the rule*

$$(gN)(hN) := (gh)N$$

*is well defined and makes  $G/N$  a group. Its identity is  $N = eN$ , and  $(gN)^{-1} = g^{-1}N$ . The canonical projection*

$$\pi : G \longrightarrow G/N, \quad \pi(g) = gN,$$

*is a surjective homomorphism with kernel  $N$ .*

*Proof.* Choose other representatives  $g' = gn_1$  and  $h' = hn_2$ , with  $n_1, n_2 \in N$ . Then

$$g'h' = gn_1hn_2 = gh(h^{-1}n_1h)n_2 \in ghN,$$

since  $N$  is normal. Thus  $g'h'N = ghN$ , proving well-definedness. Associativity follows from that of  $G$ ;  $N$  is the identity and  $g^{-1}N$  is the inverse of  $gN$ . The multiplication rule says exactly that  $\pi$  is a homomorphism. Every coset has a representative, so  $\pi$  is onto, and  $\pi(g) = N$  exactly when  $g \in N$ .  $\square$

Every normal subgroup is therefore the kernel of some homomorphism. Conversely, if this coset multiplication makes  $G/H$  a group, its canonical projection is a homomorphism with kernel  $H$ , so  $H$  must be normal. For finite  $G$ ,

$$|G/N| = [G : N] = \frac{|G|}{|N|}.$$

The quotient retains the distinctions between cosets and identifies elements within each coset. Its elements are cosets, not selected elements of  $G$ . In particular, a quotient need not be isomorphic to a subgroup of  $G$ .

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*Example 6.9* (Integers modulo  $N$ ). For a positive integer  $N$ , the subgroup  $N\mathbb{Z}$  is normal in the additive group  $\mathbb{Z}$ . The quotient has cosets

$$N\mathbb{Z}, 1 + N\mathbb{Z}, \dots, (N-1) + N\mathbb{Z},$$

and addition is  $(a + N\mathbb{Z}) + (b + N\mathbb{Z}) = (a + b) + N\mathbb{Z}$ . These are exactly the residue classes of §2.3:

$$\mathbb{Z}/N\mathbb{Z} \cong \mathbb{Z}_N.$$

This explains the quotient notation introduced there. For  $N > 1$  it also illustrates the preceding distinction:  $\mathbb{Z}$  has no nontrivial finite subgroup, although this quotient is finite.

*Example 6.10* (Quotients of the square group). Write  $D_4 = \langle r, s \mid r^4 = s^2 = 1, srs^{-1} = r^{-1} \rangle$ . Its nontrivial proper normal subgroups and quotients are

| $N$                            | $ N $ | $D_4/N$        |
|--------------------------------|-------|----------------|
| $\langle r \rangle$            | 4     | $\mathbb{Z}_2$ |
| $\langle r^2, s \rangle$       | 4     | $\mathbb{Z}_2$ |
| $\langle r^2, rs \rangle$      | 4     | $\mathbb{Z}_2$ |
| $\langle r^2 \rangle = Z(D_4)$ | 2     | $V_4$          |

The first three are normal because their index is two. For  $N = \langle r^2, s \rangle$ , the two cosets are

$$N = \{e, r^2, s, r^2s\}, \quad rN = \{r, r^3, rs, r^3s\}.$$

The coset  $N$  is the identity, and  $(rN)^2 = r^2N = N$ . Thus

| $\cdot$ | $N$  | $rN$ |
|---------|------|------|
| $N$     | $N$  | $rN$ |
| $rN$    | $rN$ | $N$  |

This is the multiplication table of  $\mathbb{Z}_2$ . The cases  $N = \langle r \rangle$  and  $N = \langle r^2, rs \rangle$  have the same table, with nonidentity cosets  $sN$  and  $rN$ , respectively.

For  $Z = Z(D_4) = \langle r^2 \rangle$ , the four cosets are

$$\begin{aligned} Z &= \{e, r^2\}, & rZ &= \{r, r^3\}, \\ sZ &= \{s, r^2s\}, & rsZ &= \{rs, r^3s\}. \end{aligned}$$

Each nonidentity coset squares to  $Z$ , since  $r^2 \in Z$  and  $s^2 = (rs)^2 = e$ . Moreover,

$$(rZ)(sZ) = rsZ, \quad (sZ)(rZ) = srZ = r^{-1}sZ = rsZ.$$

Using  $(gZ)(hZ) = (gh)Z$  gives the full table

| $\cdot$ | $Z$   | $rZ$  | $sZ$  | $rsZ$ |
|---------|-------|-------|-------|-------|
| $Z$     | $Z$   | $rZ$  | $sZ$  | $rsZ$ |
| $rZ$    | $rZ$  | $Z$   | $rsZ$ | $sZ$  |
| $sZ$    | $sZ$  | $rsZ$ | $Z$   | $rZ$  |
| $rsZ$   | $rsZ$ | $sZ$  | $rZ$  | $Z$   |

Hence  $D_4/Z(D_4) \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \cong V_4$ . The four reflection subgroups of order two are not normal: conjugating  $\langle r^k s \rangle$  by  $r$  gives  $\langle r^{k+2} s \rangle$ .

**Example 6.11** (The quaternion quotient). The group  $Q$  of Example 2.25 has centre  $Z = Z(Q) = \{1, -1\}$ . The quotient consists of

$$Z, \quad iZ = \{\pm i\}, \quad jZ = \{\pm j\}, \quad kZ = \{\pm k\}.$$

Because  $i^2 = j^2 = k^2 = -1 \in Z$ , its three nonidentity elements have order two. Further,  $ij = k$  and  $ji = -k$  give the same coset. Explicitly,

$$Z \mapsto (0, 0), \quad iZ \mapsto (1, 0), \quad jZ \mapsto (0, 1), \quad kZ \mapsto (1, 1)$$

respects multiplication, with addition modulo two on the right. Hence

$$Q/Z(Q) \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \cong V_4 \cong D_2.$$

The quotient has lost the signs that distinguished  $ij$  from  $ji$ . An Abelian quotient therefore does not imply an Abelian original group.

**Quiz 6.2.** For  $H = \langle (12) \rangle \subset S_3$ , the elements  $e$  and  $(12)$  represent the same coset  $H$ . Show that  $e(13)$  and  $(12)(13)$  represent different left cosets. Explain why multiplication of cosets fails.

In the homogeneous-space example  $S^2 \cong SO(3)/SO(2)$ , does this notation make the sphere a quotient *group*? Consider what conjugation by a rotation does to the axis fixed by  $SO(2)$ .

### 6.3. THE ISOMORPHISM THEOREMS

READING. [Moore, §7.4, pp. 99–101].

**Theorem 6.12** (First isomorphism theorem). *For a homomorphism  $\varphi : G \rightarrow G'$  with kernel  $K$ , the map*

$$\bar{\varphi} : G/K \longrightarrow \text{im } \varphi, \quad gK \longmapsto \varphi(g),$$

*is an isomorphism.*

*Proof.* If  $gK = hK$ , then  $h^{-1}g \in K$ , so  $\varphi(g) = \varphi(h)$ ; the map is well defined. Moreover,

$$\bar{\varphi}((gK)(hK)) = \varphi(gh) = \bar{\varphi}(gK)\bar{\varphi}(hK).$$

It is onto the image by definition. If  $\bar{\varphi}(gK) = e'$ , then  $g \in K$  and  $gK = K$ , so its kernel is trivial and it is injective.  $\square$

All elements in one coset of  $K = \ker \varphi$  have the same image under  $\varphi$ , and different cosets have different images. Thus  $\bar{\varphi}$  gives a bijection between the kernel cosets in  $G/K$  and the elements of  $\text{im } \varphi$ .

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*Example 6.13* (Inner automorphisms). The homomorphism  $I : G \rightarrow \text{Aut}(G)$  from §4.4, given by  $I_g(x) = gxg^{-1}$ , has kernel  $Z(G)$  and image  $\text{Inn}(G)$ . Therefore

$$G/Z(G) \cong \text{Inn}(G), \quad gZ(G) \mapsto I_g.$$

Central elements cannot be distinguished by their conjugation action. For the quaternion group this gives  $\text{Inn}(Q) \cong V_4$ .

*Example 6.14* (Determinants and rotations). The determinant maps below are surjective, with the kernels already identified in §6.1. Thus, for  $n \geq 1$ ,

$$\begin{aligned} GL(n, \kappa)/SL(n, \kappa) &\cong \kappa^\times, \\ U(n)/SU(n) &\cong U(1), \\ O(n)/SO(n) &\cong \mathbb{Z}_2. \end{aligned}$$

Surjectivity follows by using diagonal matrices with the desired determinant. For  $n \geq 2$ , the sign map similarly gives  $S_n/A_n \cong \mathbb{Z}_2$ . Theorem 3.16 now takes the quotient form

$$SU(2)/\{\pm 1\} \cong SO(3).$$

This uses the matrix homomorphism proved in Chapter 3.

Two further useful forms of the theorem describe restriction to a subgroup and taking quotients in stages. Their proofs are applications of the first theorem.

**Theorem 6.15** (Second isomorphism theorem). *If  $H \subset G$  is a subgroup and  $N \triangleleft G$ , then  $HN$  is a subgroup,  $N \triangleleft HN$ ,  $H \cap N \triangleleft H$ , and*

$$H/(H \cap N) \cong HN/N.$$

*Proof.* Let  $\pi : G \rightarrow G/N$  be the projection. Its image  $\pi(H)$  is a subgroup and its preimage is  $\pi^{-1}(\pi(H)) = HN$ . This preimage is a subgroup by the subgroup test: if  $\pi(x), \pi(y) \in \pi(H)$ , then  $\pi(xy^{-1}) = \pi(x)\pi(y)^{-1} \in \pi(H)$ . The restriction  $\pi|_H$  has kernel  $H \cap N$  and image  $\pi(H) = HN/N$ . Apply the first theorem. Normality of  $N$  in  $HN$  follows from normality in  $G$ .  $\square$

*Example 6.16.* For  $G = S_3$ ,  $N = A_3$ , and  $H = \langle (1\ 2) \rangle$ , we have  $H \cap N = \{e\}$  and  $HN = S_3$ . Thus  $S_3/A_3 \cong H \cong \mathbb{Z}_2$ . Here the quotient can be represented by a subgroup; this is an extra property, not part of the definition of a quotient.

**Theorem 6.17** (Third isomorphism theorem). *If  $N \subset H \subset G$  with both  $N$  and  $H$  normal in  $G$ , then  $H/N \triangleleft G/N$  and*

$$(G/N)/(H/N) \cong G/H.$$

*Proof.* The map  $G/N \rightarrow G/H$ ,  $gN \mapsto gH$ , is well defined because  $N \subset H$ . It is a surjective homomorphism with kernel  $H/N$ . Apply the kernel-normality result and the first isomorphism theorem.  $\square$

**Quiz 6.3.** If  $G/Z(G)$  is cyclic, prove that  $G$  is Abelian. Hint: if  $gZ(G)$  generates the quotient, every element of  $G$  is  $g^k z$  for some  $z \in Z(G)$ .

Use the third theorem with  $4\mathbb{Z} \subset 2\mathbb{Z} \subset \mathbb{Z}$  to identify  $\mathbb{Z}_4/\{\bar{0}, \bar{2}\}$ . Explain each map rather than only comparing group orders.

## 6.4. EXACT SEQUENCES AND GROUP EXTENSIONS

READING. [Moore, §7.4, pp. 98–103, §14, pp. 408–415].

### 6.4.1. Short exact sequences and group extensions

*Example 6.18* (Forgetting the translation). Return to the spatial transformations of Example 4.6. Restricting  $R$  to  $SO(3)$  gives the orientation-preserving Euclidean group, denoted  $SE(3)$ . Its elements act by  $x \mapsto Rx + a$  and multiply as

$$\{R \mid a\}\{S \mid b\} = \{RS \mid a + Rb\}.$$

The map  $\pi(\{R \mid a\}) = R$  keeps the rotation and forgets the translation. It is onto  $SO(3)$ , and its kernel is the subgroup of pure translations  $\{1 \mid a\}$ , isomorphic to  $(\mathbb{R}^3, +)$ . Hence

$$SE(3)/\mathbb{R}^3 \cong SO(3).$$

The notation  $\mathbb{R}^3$  in this quotient means its embedded translation subgroup. The information can be displayed together as

$$1 \longrightarrow \mathbb{R}^3 \xrightarrow{\iota} SE(3) \xrightarrow{\pi} SO(3) \longrightarrow 1, \quad (6.1)$$

where  $\iota(a) = \{1 \mid a\}$ . Every rotation occurs, and the transformations mapped to the identity rotation are exactly the translations. Including reflections gives the analogous sequence with  $E(3)$  and  $O(3)$ .

exact sequence  
正合列

**Definition 6.19** (Exact sequence). A sequence of groups and homomorphisms has the form

$$\cdots \longrightarrow G_{i-1} \xrightarrow{f_{i-1}} G_i \xrightarrow{f_i} G_{i+1} \longrightarrow \cdots,$$

where each  $f_i : G_i \rightarrow G_{i+1}$  is a homomorphism. It is *exact* at  $G_i$  if the image of the incoming map equals the kernel of the outgoing map,

$$\text{im } f_{i-1} = \ker f_i.$$

The sequence is *exact* if this condition holds at every intermediate group. A *short exact sequence* has the form

$$1 \longrightarrow N \xrightarrow{\iota} E \xrightarrow{\pi} G \longrightarrow 1 \quad (6.2)$$

and is exact at  $N$ ,  $E$ , and  $G$ . Here  $1$  denotes the trivial group.

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The map from 1 has image  $\{e_N\}$ , so exactness at  $N$  means  $\ker \iota = \{e_N\}$ :  $\iota$  is injective. The map to 1 has kernel all of  $G$ , so exactness at  $G$  means  $\pi$  is onto. Exactness at  $E$  is the condition  $\text{im } \iota = \ker \pi$ . Thus the short exact sequence (6.2) gives

$$N \cong \iota(N) \triangleleft E, \quad E/\iota(N) \cong G.$$

The quotient isomorphism follows from the first isomorphism theorem (Theorem 6.12) applied to  $\pi$ , since  $\ker \pi = \iota(N)$  and  $\text{im } \pi = G$ . For sequences of Abelian groups, we use additive notation and denote the trivial group by 0.

**Definition 6.20** (Group extension). A short exact sequence of the form (6.2) is an *extension of  $G$  by  $N$* . We call  $N$  its kernel group and  $G$  its quotient group. We usually identify  $N$  with its image  $\iota(N) \subset E$ .

The sequence describes which information is forgotten by  $\pi$ . The extension problem asks how that information enters multiplication in  $E$ . Specifying the abstract groups  $N$  and  $G$  alone need not determine  $E$ . For finite groups it determines  $|E| = |N||G|$ , but not its group law.

*Example 6.21* (The same kernel and quotient). Both of the following sequences are exact:

$$\begin{aligned} 0 \longrightarrow \mathbb{Z}_2 &\xrightarrow{a \mapsto 2a} \mathbb{Z}_4 \xrightarrow{b \mapsto b \bmod 2} \mathbb{Z}_2 \longrightarrow 0, \\ 0 \longrightarrow \mathbb{Z}_2 &\xrightarrow{a \mapsto (a,0)} \mathbb{Z}_2 \times \mathbb{Z}_2 \xrightarrow{(a,b) \mapsto b} \mathbb{Z}_2 \longrightarrow 0. \end{aligned}$$

The middle groups are not isomorphic:  $\mathbb{Z}_4$  has an element of order four, whereas every nonidentity element of  $\mathbb{Z}_2 \times \mathbb{Z}_2$  has order two.

### 6.4.2. Split extensions and semidirect products

**Definition 6.22** (Splitting). The short exact sequence (6.2) *splits* if

$$\exists \text{ homomorphism } s : G \rightarrow E \quad \text{with} \quad \pi \circ s = \text{id}_G.$$

The map  $s$  chooses one representative of each quotient element *while preserving multiplication*. Choosing representatives alone does not ensure this. For the sequence (6.1), choosing an origin gives  $s(R) = \{R \mid 0\}$ , which is a homomorphism, so that extension splits.

In the  $\mathbb{Z}_4$  sequence, the projection is  $\pi(x) = x \bmod 2$ . The nonzero element  $1 \in \mathbb{Z}_2$  has two preimages, 1 and 3 in  $\mathbb{Z}_4$ , each of order four. A splitting must choose one of these as  $s(1)$ . Since  $1 + 1 = 0$  in  $\mathbb{Z}_2$ , a homomorphism would require

$$s(1) + s(1) = s(1 + 1) = s(0) = 0.$$

But  $1 + 1 = 2$  and  $3 + 3 = 2$  in  $\mathbb{Z}_4$ . Thus neither choice preserves addition, and the extension does not split.

For the  $\mathbb{Z}_2 \times \mathbb{Z}_2$  sequence, the projection is  $\pi(a, b) = b$ . Choose  $s(b) = (0, b)$ . Then

$$\begin{aligned}\pi(s(b)) &= b, \\ s(b + b') &= (0, b + b') = (0, b) + (0, b') = s(b) + s(b').\end{aligned}$$

Thus  $s$  is a homomorphism with  $\pi \circ s = \text{id}_{\mathbb{Z}_2}$ , so this extension splits. Here  $s(1) = (0, 1)$  has order two.

**Theorem 6.23** (Structure of a split extension). *Suppose the short exact sequence (6.2) splits with  $s : G \rightarrow E$ . Identify  $N$  with its image  $\iota(N) \triangleleft E$ . Conjugation by  $s(g)$  defines*

$$\alpha_g : N \longrightarrow N, \quad \alpha_g(n) = s(g)ns(g)^{-1}.$$

*Then  $\alpha : G \rightarrow \text{Aut}(N)$ ,  $g \mapsto \alpha_g$ , is a homomorphism. Every  $x \in E$  has a unique expression  $x = \iota(n)s(g)$ , with  $n \in N$  and  $g \in G$ . The product is taken in  $E$  and is abbreviated  $ns(g)$  after identifying  $N$  with  $\iota(N)$ . Thus the map*

$$\Phi : N \times G \longrightarrow E, \quad (n, g) \longmapsto \iota(n)s(g),$$

*is a bijection of sets. Under this bijection, multiplication in  $E$  becomes*

$$(n, g)(m, h) = (n\alpha_g(m), gh). \quad (6.3)$$

*With this multiplication,  $N \times G$  is a group and  $\Phi$  is an isomorphism. The factorization  $(n, g) = (n, e_G)(e_N, g)$  corresponds to  $\iota(n)s(g)$  in  $E$ .*

*Proof.* Conjugation by  $s(g)$  maps the normal subgroup  $N$  onto itself, with inverse given by conjugation by  $s(g)^{-1}$ . Thus  $\alpha_g \in \text{Aut}(N)$ . Since  $s(gh) = s(g)s(h)$ ,

$$\begin{aligned}\alpha_{gh}(n) &= s(g)s(h)ns(h)^{-1}s(g)^{-1} \\ &= \alpha_g(\alpha_h(n)).\end{aligned}$$

Hence  $\alpha_{gh} = \alpha_g \circ \alpha_h$ , proving that  $\alpha : G \rightarrow \text{Aut}(N)$  is a homomorphism.

For  $x \in E$ , put  $g = \pi(x)$ . Then  $xs(g)^{-1} \in \ker \pi = \iota(N)$ , giving  $x = \iota(n)s(g)$ . If  $\iota(n)s(g) = \iota(m)s(h)$ , applying  $\pi$  gives  $g = h$ , then injectivity of  $\iota$  gives  $n = m$ . Thus  $\Phi$  is a bijection. Finally, using  $s(g)\iota(m)s(g)^{-1} = \iota(\alpha_g(m))$ ,

$$\begin{aligned}\Phi(n, g)\Phi(m, h) &= \iota(n)s(g)\iota(m)s(h) \\ &= \iota(n\alpha_g(m))s(gh) \\ &= \Phi(n\alpha_g(m), gh).\end{aligned}$$

Hence multiplication in  $E$  gives precisely the pair multiplication in (6.3). Transporting the group law through the bijection  $\Phi$  makes  $N \times G$  a group isomorphic to  $E$ .  $\square$

semidirect product  
半直积

**Definition 6.24** (Semidirect product). For any homomorphism  $\alpha : G \rightarrow \text{Aut}(N)$ , write  $\alpha_g = \alpha(g)$  for the automorphism of  $N$  assigned to  $g$ . The *semidirect product*  $N \rtimes_\alpha G$  is the set  $N \times G$  of ordered pairs  $(n, g)$ , with  $n \in N$  and  $g \in G$ , equipped with the multiplication (6.3).

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The identity is  $(e_N, e_G)$  and

$$(n, g)^{-1} = (\alpha_{g^{-1}}(n^{-1}), g^{-1}).$$

Since  $\alpha_{gh} = \alpha_g \circ \alpha_h$ , either bracketing of  $(n, g)(m, h)(\ell, k)$  gives

$$(n\alpha_g(m)\alpha_{gh}(\ell), ghk),$$

which proves associativity. The maps  $\iota(n) = (n, e_G)$  and  $\pi(n, g) = g$  give the short exact sequence

$$1 \longrightarrow N \xrightarrow{\iota} N \rtimes_{\alpha} G \xrightarrow{\pi} G \longrightarrow 1.$$

It splits with  $s(g) = (e_N, g)$ . Thus every split extension is isomorphic to a semidirect product, and every semidirect product gives a split extension. If every  $\alpha_g$  is the identity map of  $N$ , this is the direct product  $N \times G$ .

*Example 6.25* (Rotations act on translations). In the spatial example, conjugation gives

$$\{R \mid 0\}\{\mathbb{1} \mid b\}R^{-1} \mid 0 = \{\mathbb{1} \mid Rb\}.$$

The action in the semidirect product is therefore  $\alpha_R(b) = Rb$ , and  $SE(3) \cong \mathbb{R}^3 \rtimes SO(3)$ . A translation direction changes under rotation, which explains the term  $Rb$  in the product. The same calculation on the square lattice recovers Example 4.23:

$$p4mm \cong \mathbb{Z}^2 \rtimes D_4.$$

Here every point-group operation has a representative with zero translation, and these representatives form a subgroup.

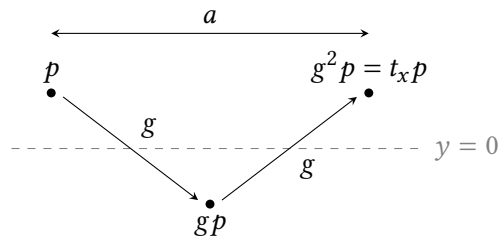
### 6.4.3. Nonsplit extensions and glide reflections

*Example 6.26* (A nonsplit spatial extension). Take  $a, b > 0$  and let  $\Gamma$  be the group generated on the plane by

$$t_x(x, y) = (x + a, y), \quad t_y(x, y) = (x, y + b), \quad g(x, y) = (x + a/2, -y).$$

The operation  $g$  is a glide reflection. Direct composition gives

$$g^2 = t_x, \quad gt_xg^{-1} = t_x, \quad gt_yg^{-1} = t_y^{-1}.$$



Consequently every element has the form  $t_x^m t_y^n$  or  $t_x^m t_y^n g$ , with  $m, n \in \mathbb{Z}$ . Their linear parts are respectively 1 and the reflection  $M(x, y) = (x, -y)$ . The full translation subgroup is  $T = \langle t_x, t_y \rangle \cong \mathbb{Z}^2$ , and forgetting translations gives

$$1 \longrightarrow T \longrightarrow \Gamma \xrightarrow{\pi} \{1, M\} \longrightarrow 1.$$

The quotient reflection has order two. However, every representative of it has square

$$(t_x^m t_y^n g)^2 = t_x^{2m+1} \neq e.$$

No representative has order two, so no homomorphism  $s$  can split this sequence. A point-group reflection need not be an actual reflection in this group: here it must come with half a translation.

A space group is *symmorphic* if and only if its point-group projection admits a homomorphic choice of representatives for the whole point group, so the extension by its full translation lattice splits. If no such choice exists, it is *nonsymmorphic*, as in this glide example. The translation and point groups alone therefore do not determine the space group.

#### 6.4.4. Central extensions

central extension  
中心扩张

**Definition 6.27** (Central extension). An extension

$$1 \longrightarrow A \longrightarrow E \xrightarrow{\pi} G \longrightarrow 1$$

is *central* if its kernel subgroup  $A$  lies in  $Z(E)$ .

The kernel group  $A$  is then Abelian, but  $E$  need not be. For example,  $Q$  is a central extension of  $V_4$  by  $\{\pm 1\}$ , as Example 6.11 shows.

Centrality and splitting are different conditions. Centrality asks whether the kernel commutes with every element; splitting asks whether quotient elements have representatives that multiply consistently. If a central extension splits, the action  $\alpha$  in Theorem 6.23 is trivial, so  $E \cong A \times G$ . The central extension with middle group  $\mathbb{Z}_4$  does not split.

**Quiz 6.4.** Write  $D_n \cong \mathbb{Z}_n \rtimes \mathbb{Z}_2$  explicitly: which automorphism of  $\mathbb{Z}_n$  does the nonidentity element of  $\mathbb{Z}_2$  induce? For which  $n \geq 2$  does this become a direct product?

*Optional.* Verify  $\varphi I_g \varphi^{-1} = I_{\varphi(g)}$  for  $\varphi \in \text{Aut}(G)$ . Deduce  $\text{Inn}(G) \triangleleft \text{Aut}(G)$ , define  $\text{Out}(G) = \text{Aut}(G)/\text{Inn}(G)$ , and check exactness of

$$1 \longrightarrow Z(G) \longrightarrow G \xrightarrow{I} \text{Aut}(G) \longrightarrow \text{Out}(G) \longrightarrow 1.$$

## 6. Normal subgroups, quotients, and extensions

### 6.5. THE FINITE HEISENBERG GROUP

Central extensions, introduced in §6.4, occur naturally in quantum mechanics. Position and momentum displacement operators give an extension of the additive displacement group  $(\mathbb{R}^2, +)$  by the scalar phase group  $U(1)$ . The finite Heisenberg group gives a discrete version of this construction.

#### 6.5.1. Position and momentum displacement operators

READING. [Moore, §7.4, pp. 102–103, §11.17.2, pp. 314–317].

The position and momentum operators act on a wavefunction  $\psi(x)$  on the line by

$$(\hat{x}\psi)(x) = x\psi(x), \quad (\hat{p}\psi)(x) = -i\hbar \frac{d\psi}{dx}.$$

Their operator commutator is  $[\hat{x}, \hat{p}] = \hat{x}\hat{p} - \hat{p}\hat{x}$ . The product rule gives  $[\hat{x}, \hat{p}] = i\hbar 1$ . A plane wave with wave number  $k$  and momentum  $p_0 = \hbar k$  is

$$\psi_{p_0}(x) = e^{ip_0 x/\hbar}, \quad \hat{p}\psi_{p_0} = p_0\psi_{p_0}.$$

A displacement  $a$  in position and a displacement  $b$  in momentum are implemented by the operators  $T_a$  and  $M_b$ ,

$$(T_a\psi)(x) = \psi(x - a), \quad (M_b\psi)(x) = e^{ibx/\hbar}\psi(x).$$

These operations are generated by momentum and position, respectively, and are written  $T_a = e^{-ia\hat{p}/\hbar}$  and  $M_b = e^{ib\hat{x}/\hbar}$ . Their effects on a plane wave are

$$\begin{aligned} (T_a\psi_{p_0})(x) &= e^{-iap_0/\hbar}\psi_{p_0}(x), \\ (M_b\psi_{p_0})(x) &= e^{i(p_0+b)x/\hbar} = \psi_{p_0+b}(x). \end{aligned}$$

Thus  $M_b$  increases the momentum label by  $b$ . Translating a plane wave by  $a$  gives a scalar phase that depends on its momentum. The two orders therefore give

$$\begin{aligned} M_b T_a \psi_{p_0} &= e^{-iap_0/\hbar} \psi_{p_0+b}, \\ T_a M_b \psi_{p_0} &= e^{-ia(p_0+b)/\hbar} \psi_{p_0+b}. \end{aligned}$$

The same phase difference follows for any wavefunction from the coordinate formulas above, giving the Weyl relation

$$M_b T_a = e^{iab/\hbar} T_a M_b.$$

To identify the extension explicitly, let  $\mathcal{W}$  be the group of displacement operators with scalar phases,

$$\mathcal{W} = \{zT_a M_b : z \in U(1), a, b \in \mathbb{R}\}.$$

The Weyl relation gives its multiplication law,

$$(zT_a M_b)(wT_{a'} M_{b'}) = zw e^{iba'/\hbar} T_{a+a'} M_{b+b'}.$$

The displacement labels  $a, b$  and the phase  $z$  are uniquely determined by the action on wavefunctions. A localized wavefunction determines  $a$  from the shift

of its support. After cancelling  $T_a$ , equality of the multiplication factors  $ze^{ibx/\hbar}$  for every  $x$  determines  $b$  and  $z$ . The projection  $\pi(zT_aM_b) = (a, b)$  maps onto the displacement group  $(\mathbb{R}^2, +)$ , whose multiplication is addition of pairs. The displayed product shows that  $\pi$  is a homomorphism. Its kernel is exactly the subgroup  $\{z\mathbb{1} : z \in U(1)\}$ , which commutes with every element of  $\mathcal{W}$ . Thus we have the central extension

$$1 \longrightarrow U(1) \xrightarrow{z \mapsto z\mathbb{1}} \mathcal{W} \xrightarrow{\pi} (\mathbb{R}^2, +) \longrightarrow 1.$$

This is an extension of the displacement group  $(\mathbb{R}^2, +)$  by the phase group  $U(1)$ . In  $\mathcal{W}/U(1)$ , a coset consists of all operators  $zT_aM_b$  with the same pair  $(a, b)$ . The two products  $M_bT_a$  and  $T_aM_b$  lie in that same coset, because their ratio is the scalar phase  $e^{iab/\hbar}\mathbb{1}$  in the kernel. Consequently their quotient elements commute.

### 6.5.2. Clock and shift matrices

For a finite analogue, take  $N \geq 2$  equally spaced sites on a ring of circumference  $L$ , at  $x_j = jL/N$ , with periodic labels  $j \in \mathbb{Z}_N$ . The amplitudes  $\psi_j$  form a vector  $\psi \in \mathbb{C}^N$ . Choose  $a = L/N$  and  $b = 2\pi\hbar/L$ , so the phase above is  $\omega = e^{2\pi i/N}$ . The analogues of  $T_a$  and  $M_b$  are the matrices  $P$  and  $Q$  defined by

$$(P\psi)_j = \psi_{j-1}, \quad (Q\psi)_j = \omega^j \psi_j. \quad (6.4)$$

The operator  $P$  moves the amplitudes by one site, while  $Q$  applies a position-dependent phase. The choice of phases in  $Q$  shifts a discrete plane wave  $\psi_j = \omega^{kj}$  to  $\omega^{(k+1)j}$ , increasing its wave label by one. Thus  $P$  shifts the site label and  $Q$  shifts the wave label. Both operations preserve  $\sum_j |\psi_j|^2$ .

In the basis  $e_0, \dots, e_{N-1}$ , these are the *shift* and *clock* matrices:

$$Pe_j = e_{j+1}, \quad Qe_j = \omega^j e_j,$$

where labels are modulo  $N$ . For  $N = 3$ ,

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \omega & 0 \\ 0 & 0 & \omega^2 \end{pmatrix}.$$

Both have order  $N$ . Their products satisfy

$$(QP\psi)_j = \omega^j \psi_{j-1}, \quad (PQ\psi)_j = \omega^{j-1} \psi_{j-1},$$

hence the *Weyl relation*

$$QP = \omega PQ, \quad QPQ^{-1}P^{-1} = \omega\mathbb{1}. \quad (6.5)$$

Both  $QP$  and  $PQ$  contain one site shift and one wave-label shift, so they give the same pair of displacement counts  $(1, 1)$ . Their matrix products satisfy  $QP = \omega PQ$ . Any group containing  $P$  and  $Q$  must also contain this scalar phase, since  $QPQ^{-1}P^{-1} = \omega\mathbb{1}$ .

Weyl relation  
外尔关系

Heisenberg group  
海森堡群

## 6. Normal subgroups, quotients, and extensions

**Definition 6.28** (Finite Heisenberg group). The *finite Heisenberg group* is the matrix group

$$\text{Heis}_N = \langle P, Q, \omega \mathbb{1} \rangle \subset U(N).$$

**Proposition 6.29.** Every element of  $\text{Heis}_N$  has a unique form  $\omega^a P^b Q^c$ , with  $a, b, c \in \mathbb{Z}_N$ . Consequently its order is  $N^3$ . Writing an element as  $(a, b, c)$ , the multiplication is

$$(a, b, c)(a', b', c') = (a + a' + cb', b + b', c + c'), \quad (6.6)$$

with all entries reduced modulo  $N$ .

*Proof.* The relation  $Q^c P^{b'} = \omega^{cb'} P^{b'} Q^c$  moves every  $P$  to the left of every  $Q$ , proving existence and the multiplication formula. For uniqueness, observe

$$\omega^a P^b Q^c e_j = \omega^{a+cj} e_{j+b}.$$

The image of  $e_0$  determines  $b$  and  $a$ ; the image of  $e_1$  then determines  $c$ , since  $\omega$  is a primitive  $N$ th root of unity.  $\square$

**Proposition 6.30** (The central extension). The centre of  $\text{Heis}_N$  is  $\{\omega^a \mathbb{1} : a \in \mathbb{Z}_N\} \cong \mathbb{Z}_N$ , and there is a nonsplit central extension

$$1 \longrightarrow \mathbb{Z}_N \xrightarrow{a \mapsto \omega^a \mathbb{1}} \text{Heis}_N \xrightarrow{\pi} \mathbb{Z}_N \times \mathbb{Z}_N \longrightarrow 1, \quad (6.7)$$

Here  $\pi(\omega^a P^b Q^c) = (b, c)$  records the number  $b$  of site shifts and the number  $c$  of wave-label shifts, modulo  $N$ . The kernel is the scalar phase subgroup  $\{\omega^a \mathbb{1} : a \in \mathbb{Z}_N\}$ . Thus  $\text{Heis}_N$  is a central extension of  $\mathbb{Z}_N \times \mathbb{Z}_N$  by the phase group  $\mathbb{Z}_N$ .

*Proof.* An element  $\omega^a P^b Q^c$  commutes with  $P$  exactly when  $\omega^c = 1$ , and with  $Q$  exactly when  $\omega^b = 1$ . Thus a central element has  $b = c = 0$ ; all remaining scalar matrices are indeed central. Formula (6.6) shows that  $\pi$  is a surjective homomorphism with this centre as kernel, proving exactness.

For a splitting, the commuting elements  $(1, 0)$  and  $(0, 1)$  would need commuting lifts. Every such pair of lifts has the form  $\omega^r P$  and  $\omega^s Q$ . Multiplying by these scalar phases does not change the relation  $QP = \omega PQ$ , so they cannot commute. No splitting exists.  $\square$

**Example 6.31** (The two-site case). For  $N = 2$ ,  $\omega = -1$ , and  $P = X, Q = Z$  are the Pauli matrices of §2.7. Thus

$$\text{Heis}_2 = \{\pm \mathbb{1}, \pm X, \pm Z, \pm XZ\}.$$

This is an eight-element subgroup of the sixteen-element Pauli group  $P_1$  used there; adjoining  $i\mathbb{1}$  gives  $P_1$ . In particular,  $XZ = -ZX$  and  $(XZ)^2 = -1$ , even though  $X^2 = Z^2 = \mathbb{1}$ . The orders of the generators do not determine the orders of their products.

## 6.5.3. The discrete Fourier basis

For  $k \in \mathbb{Z}_N$ , define the discrete plane wave

$$\psi_j^{(k)} = \frac{1}{\sqrt{N}} \omega^{kj}.$$

These  $N$  vectors form an orthonormal basis, since the geometric series gives

$$\langle \psi^{(k)}, \psi^{(\ell)} \rangle = \frac{1}{N} \sum_{j=0}^{N-1} \omega^{(\ell-k)j} = \delta_{k\ell}.$$

Directly,

$$P\psi^{(k)} = \omega^{-k}\psi^{(k)}, \quad Q\psi^{(k)} = \psi^{(k+1)}.$$

In the site basis,  $P$  shifts the label and  $Q$  is diagonal. In the wave basis,  $P$  is diagonal and  $Q$  shifts the wave label. This is the finite Fourier transform, and in the particle interpretation  $k$  labels discrete momentum modulo  $N$ . The extra phase in (6.5) records how these two kinds of displacement combine. These operators act on states. They are symmetries of a Hamiltonian when they commute with it.

The finite construction preserves the exponentiated displacement relation. The canonical commutator itself requires infinite-dimensional operators, since the trace of any finite matrix commutator is zero, whereas  $\text{tr}(i\hbar \mathbb{1}) = i\hbar N$ .

## 6.5.4. Projective representations and central extensions

A pure quantum state is represented by a *ray*, a one-dimensional complex subspace represented by any of its nonzero vectors. Normalized vectors  $\psi$  and  $e^{i\theta}\psi$  represent the same ray. If two unitary operators satisfy  $V = e^{i\theta}U$ , with one scalar phase multiplying the whole matrix, then  $V\psi = e^{i\theta}U\psi$  for every  $\psi$ . Thus they induce the same map on all rays. If  $G$  acts on these rays and unitary operators  $U(g)$  implement that action, their products can satisfy

$$U(g)U(h) = c(g, h)U(gh), \quad c(g, h) \in U(1). \quad (6.8)$$

The phase is independent of  $\psi$ : two linear invertible operators inducing the same transformation on every ray differ by a scalar. To see this, apply their ratio to two independent vectors and to their sum; linearity forces their scalar factors to agree. Unitarity makes that common scalar have modulus one. In dimension one the claim is immediate.

Equation (6.8) is called a *projective representation*. With  $U(e) = 1$ , associativity imposes

$$c(g, h)c(gh, k) = c(h, k)c(g, hk), \quad c(e, g) = c(g, e) = 1. \quad (6.9)$$

This follows simply by expanding  $(U(g)U(h))U(k) = U(g)(U(h)U(k))$ . It lets us retain the phase as an extra coordinate: on  $U(1) \times G$ , put

$$(z, g)(w, h) = (zwc(g, h), gh).$$

## 6. Normal subgroups, quotients, and extensions

The identity is  $(1, e)$  and the inverse is

$$(z, g)^{-1} = (z^{-1}c(g, g^{-1})^{-1}, g^{-1}).$$

Equation (6.9) also gives  $c(g, g^{-1}) = c(g^{-1}, g)$ . Thus this is a group, denoted  $\tilde{G}$ . Projection onto  $g$  yields a central extension

$$1 \longrightarrow U(1) \longrightarrow \tilde{G} \longrightarrow G \longrightarrow 1,$$

and  $(z, g) \mapsto zU(g)$  is now an ordinary homomorphism. For the finite example, choose  $U(b, c) = P^b Q^c$ ; the multiplier is  $\omega^{cb'}$ , and the subgroup with phases in  $\{1, \omega, \dots, \omega^{N-1}\}$  is precisely  $\text{Heis}_N$ .

Changing the phase convention to  $U'(g) = b(g)U(g)$  changes the multiplier to  $c'(g, h) = b(g)b(h)b(gh)^{-1}c(g, h)$ . Sometimes one can make every multiplier equal to one, but the noncommuting lifts in (6.5) show that this is not always possible. An overall phase alone does not distinguish physical states; it is the consistent composition of their operators that requires this extra data.

*Example 6.32* (Returning to spin and rotations). The previously established map  $R : SU(2) \rightarrow SO(3)$  gives

$$1 \longrightarrow \{\pm 1\} \longrightarrow SU(2) \xrightarrow{R} SO(3) \longrightarrow 1.$$

The kernel  $\{\pm 1\}$  is central because scalar matrices commute with every matrix. Let  $r \in SO(3)$  be a rotation through  $\pi$  about the third axis, so  $r^2 = \mathbb{1}_3$ . A splitting would be a homomorphism  $s : SO(3) \rightarrow SU(2)$  satisfying  $R \circ s = \text{id}_{SO(3)}$ . The homomorphism condition would require

$$s(r)^2 = s(r^2) = s(\mathbb{1}_3) = \mathbb{1}_2.$$

But  $s(r)$  must be a lift of  $r$ , and the only two lifts are  $\pm i\sigma^3$ . Since  $(\sigma^3)^2 = \mathbb{1}_2$ , both satisfy

$$(\pm i\sigma^3)^2 = -\mathbb{1}_2.$$

Neither lift satisfies the required relation. Thus no homomorphic splitting exists, and this central extension does not split.

Allowing arbitrary scalar phases does not remove the obstruction either. The half-turns about the first and third axes commute in  $SO(3)$ , but their spinor lifts, proportional to  $\sigma^1$  and  $\sigma^3$ , anticommute. Rephasing either matrix leaves that relation unchanged. Thus the two-component spinor matrices give a representation of  $SU(2)$ , but only a projective representation of  $SO(3)$ . The spin interpretation will be developed in Part II.

**Quiz 6.5.** For  $x = (a, b, c)$  and  $y = (a', b', c')$  in  $\text{Heis}_N$ , use (6.6) to find  $x^{-1}$  and show

$$xyx^{-1}y^{-1} = \omega^{cb' - c'b} \mathbb{1}.$$

Explain why the quotient is Abelian although the group is not.

## KEY POINTS

- Normality is exactly what makes coset multiplication well defined. Normal subgroups are precisely kernels of homomorphisms.
- $G/\ker \varphi \cong \operatorname{im} \varphi$ : a homomorphism identifies exactly the elements in each kernel coset.
- An extension records a kernel and a quotient together with their maps. These two groups alone do not determine the middle group.
- A splitting gives a semidirect product. Spatial symmetries illustrate both splitting and its failure.
- Centrality does not imply splitting. Clock and shift operators retain a phase lost by their Abelian quotient; this prepares the unitary projective representations of Part II.

KEY TERMS. normal subgroup 正规子群 · quotient group 商群 · simple group 单群 · exact sequence 正合列 · group extension 群扩张 · split extension 分裂扩张 · semidirect product 半直积 · central extension 中心扩张 · Heisenberg group 海森堡群 · projective representation 射影表示